Software Requirement Specification (SRS)

E-Commerce Order Management Database System — Week 1

1. Introduction

1.1 Purpose

This SRS defines the requirements for a database system that supports the daily operations of an e-commerce company, including customer registration, product management, inventory tracking, order processing, payment management, shipment tracking, and customer reviews. It serves as the foundation for subsequent ER modeling, normalization, and SQL implementation.

1.2 Scope

The system covers the full order lifecycle: a customer registers, browses the product catalog (organized by category and sourced from suppliers), places an order composed of one or more order line items, makes a payment, receives a shipment, and may leave a review. The database must store this data centrally, minimize redundancy, and support business reporting.

1.3 Intended Audience

Database designers, developers, project reviewers/instructors, and business stakeholders who need a shared understanding of the system's data requirements before design begins.

1.4 Definitions and Acronyms

SRS — Software Requirement Specification. ER — Entity-Relationship. DBA — Database Administrator.

2. Overall Description

2.1 Product Perspective

The system is a standalone relational database supporting an e-commerce platform's back-office operations. It is built exclusively around nine mandatory core entities: Customer, Category, Product, Supplier, Order, Order Details, Payment, Shipment, and Review.

2.2 Core Entities and Attributes

Entity	Attributes
Customer	Customer ID, Name, Email, Mobile Number, Address, Registration Date
Category	Category ID, Category Name, Description
Product	Product ID, Product Name, Price, Stock Quantity, Category ID
Supplier	Supplier ID, Supplier Name, Contact Information
Order	Order ID, Customer ID, Order Date, Total Amount, Order Status
Order Details	Order Detail ID, Order ID, Product ID, Quantity, Unit Price
Payment	Payment ID, Order ID, Payment Method, Payment Date, Payment Status

Entity	Attributes
Shipment	Shipment ID, Order ID, Delivery Address, Shipment Date, Delivery Status
Review	Review ID, Customer ID, Product ID, Rating, Comments

2.3 User Classes

Customers, order/sales staff, inventory/warehouse staff, finance staff, logistics staff, customer support staff, the DBA, and management/business users (see Stakeholder Analysis in the Business Requirement Document for role details).

2.4 Operating Constraints

Only the nine predefined entities and their listed attributes may be used; no entity or attribute may be added, removed, or renamed for the duration of the project.

3. Functional Requirements

ID	Requirement
FR-01	Register a customer with Name, Email, Mobile Number, and Address.
FR-02	Create and maintain product categories.
FR-03	Create, update, and assign products to a category, including price and stock quantity.
FR-04	Record and maintain supplier details and contact information.
FR-05	Create an order for a customer and record the order date and status.
FR-06	Add one or more order detail line items (product, quantity, unit price) to an order.
FR-07	Calculate an order's Total Amount from its order details.
FR-08	Update Order Status as the order progresses (e.g., Pending to Delivered).
FR-09	Record a payment (method, date, status) against an order.
FR-10	Record a shipment (delivery address, shipment date, delivery status) against an order.
FR-11	Allow a customer to submit a rating and comment (review) for a purchased product.
FR-12	Generate reports on sales, inventory levels, order history, and customer activity.

4. Non-Functional Requirements

- **Performance:** Queries on order, product, and customer data shall return promptly under normal load.
- **Reliability/Integrity:** Referential integrity shall be enforced between related entities (e.g., Order → Customer, Order Details → Order/Product).
- **Efficiency:** The schema shall be normalized to reduce data redundancy.
- **Security:** Access to customer and payment data shall be restricted to authorized users/processes.

- **Availability:** The database shall be available during business hours with minimal downtime.
- **Scalability:** The design shall accommodate growth in customers, products, and orders.
- **Maintainability:** The schema and documentation shall be clear enough to support future changes.

5. External Interface Requirements

This phase does not define a user interface or external system integrations; the database is the interface surface, accessed via SQL by application/reporting layers to be developed in later phases.

6. Assumptions and Constraints

See Section 6 of the Business Requirement Document for the full list of assumptions and constraints governing this system, including the restriction to the nine predefined core entities.

7. Appendix — Traceability

Each functional requirement above traces directly to one or more core entities and to a stakeholder role identified in the Business Requirement Document, ensuring that the forthcoming ER model and database schema fully satisfy the analyzed business needs.